Lecture 5 — Multiple Regression: Further Issues and Qualitative Information

Introductory Econometrics — Télécom Paris

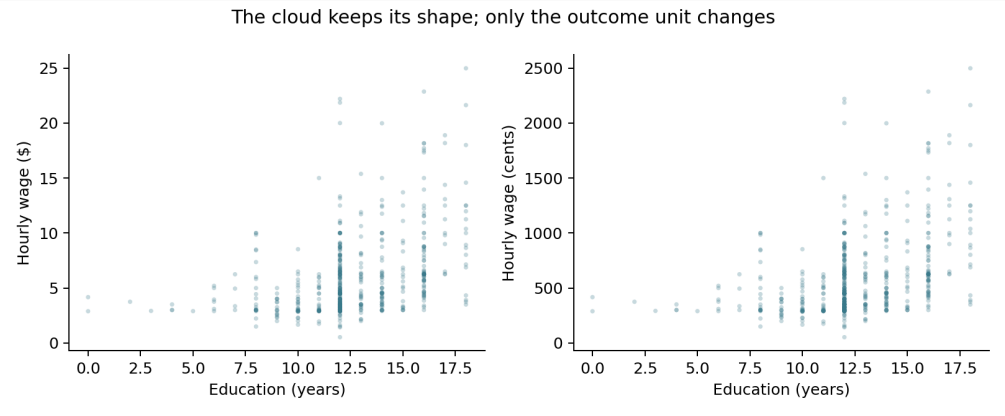
Màxim Sandiumenge i Boy

Reading: Chapters 6–7

Chapter 6
Multiple Regression Analysis: Further Issues

Effects of Data Scaling

Application: the same wage scatter in two units



Wooldridge's `wage1` data ($n = 526$ workers). The same points appear in both panels; only the wage unit changes.

Application: coefficients follow the chosen units

Regress wage on education, experience and tenure in all three cases.

	Baseline	Wage in cents	Education in decades
Wage unit	dollars	cents	dollars
Education unit	years	years	decades
Education slope	0.599	59.897	5.990
SE of slope	0.051	5.128	0.513
R^2	0.306	0.306	0.306

- Dollars to cents: the slope and its SE multiply by 100.
- Years to decades: both multiply by 10. The R^2 stays the same.

Scaling the variables in a regression

Start from the population model:

$$y_i = \beta_0 + \beta_1 x_i + u_i.$$

If $y_i^* = c_y y_i$ and $x_i^* = c_x x_i$, with $c_y, c_x > 0$, the same model in the new units is

$$y_i^* = c_y \beta_0 + \frac{c_y}{c_x} \beta_1 x_i^* + c_y u_i.$$

OLS estimates therefore satisfy

$$\hat{\beta}_0^* = c_y \hat{\beta}_0, \qquad \hat{\beta}_1^* = \frac{c_y}{c_x} \hat{\beta}_1, \qquad \hat{u}_i^* = c_y \hat{u}_i.$$

Scaling uncertainty and the test statistic

Let $k = c_y/c_x > 0$. In simple regression with an intercept, $S_{xx} = \sum_i (x_i - \bar{x})^2$ and $\text{se}(\hat{\beta}_1) = \hat{\sigma}/\sqrt{S_{xx}}$.

Notice: $\hat{\sigma}^* = \sqrt{\dfrac{\sum_i (c_y \hat{u}_i)^2}{n-2}} = c_y \hat{\sigma}, \qquad S^*_{xx} = \sum_i [c_x(x_i - \bar{x})]^2 = c_x^2 S_{xx}.$

- **Standard error:** $\text{se}(\hat{\beta}_1^*) = \dfrac{c_y \hat{\sigma}}{c_x \sqrt{S_{xx}}} = k \, \text{se}(\hat{\beta}_1)$.
- **Same substantive test:** $H_0 : \beta_1 = b$ becomes $H_0 : \beta_1^* = kb$. Thus

$$t^* = \frac{k\hat{\beta}_1 - kb}{k \, \text{se}(\hat{\beta}_1)} = t.$$

- **Confidence interval:** both endpoints of $\hat{\beta}_1 \pm t_{n-2,\,0.975} \, \text{se}(\hat{\beta}_1)$ multiply by k.

Scaling residuals and goodness of fit

Because $\hat{u}_i^* = c_y \hat{u}_i$ and $y_i^* - \bar{y}^* = c_y(y_i - \bar{y})$:

$$\mathrm{SSR}^* = c_y^2 \,\mathrm{SSR}, \qquad \mathrm{SST}^* = c_y^2 \,\mathrm{SST}\,.$$

- **Standard error of the regression:**

$$\hat{\sigma}^* = \sqrt{\frac{\mathrm{SSR}^*}{n-2}} = c_y \hat{\sigma}.$$

- **Goodness of fit:**

$$R^{2*} = 1 - \frac{\mathrm{SSR}^*}{\mathrm{SST}^*} = 1 - \frac{\mathrm{SSR}}{\mathrm{SST}} = R^2.$$

Changing only the unit of x leaves residuals, SSR and $\hat{\sigma}$ unchanged. Changing the unit of y scales these quantities as shown.

Scaling a variable inside a logarithm

If $\log y_i = \beta_0 + \beta_1 \log x_i + u_i$ and $y_i^* = c_y y_i$, then

$$\log y_i^* = (\beta_0 + \log c_y) + \beta_1 \log x_i + u_i.$$

- Every log outcome shifts vertically by $\log c_y$; the intercept absorbs it.

If instead $x_i^* = c_x x_i$, then

$$\log y_i = (\beta_0 - \beta_1 \log c_x) + \beta_1 \log x_i^* + u_i.$$

- The log regressor shifts by $\log c_x$ while $\log y$ stays fixed. The intercept adjusts to keep every fitted value unchanged. The slope and elasticity β_1 stay the same.

Standardized (beta) coefficients

Standardize y and each regressor x_j using their sample means and standard deviations:

$$z_{yi} = \frac{y_i - \bar{y}}{s_y}, \qquad z_{x_j i} = \frac{x_{ji} - \bar{x}_j}{s_{x_j}}.$$

In the regression of z_y on the standardized regressors,

$$\hat{\beta}_j^{\mathrm{std}} = \hat{\beta}_j \frac{s_{x_j}}{s_y}, \qquad \hat{\beta}_0^{\mathrm{std}} = 0.$$

The intercept is zero because the standardized outcome and regressors all have sample mean zero.

The beta coefficients help identify which regressors have larger conditional associations on a common scale. Interpret a one-standard-deviation change only when it is meaningful within that variable's observed range; a large beta alone does not establish causal importance.

Application: pollution and housing prices

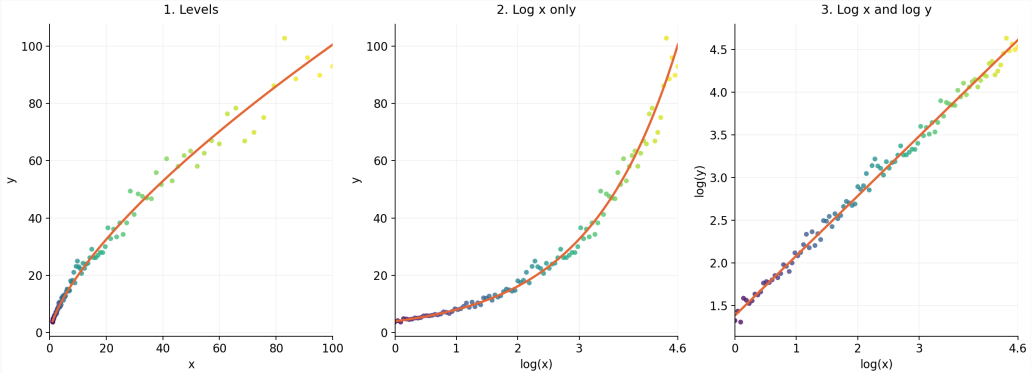
Wooldridge's Example 6.1 uses Boston-area communities: median house price, nitrogen oxide pollution (nox), crimes per capita, average rooms, distance to employment centers and student–teacher ratio. Each variable is standardized.

$$\widehat{z_{price}} = -0.340 z_{nox} - 0.143 z_{crime} + 0.514 z_{rooms} - 0.235 z_{dist} - 0.270 z_{stratio}.$$

- Holding the other characteristics fixed,
  - One SD more nitrogen oxide pollution is associated with 0.340 SD lower price.
  - One SD more crime is associated with 0.143 SD lower price.
- Rooms has the largest coefficient in absolute value.

More on Functional Form

What taking logarithms does to a scatterplot



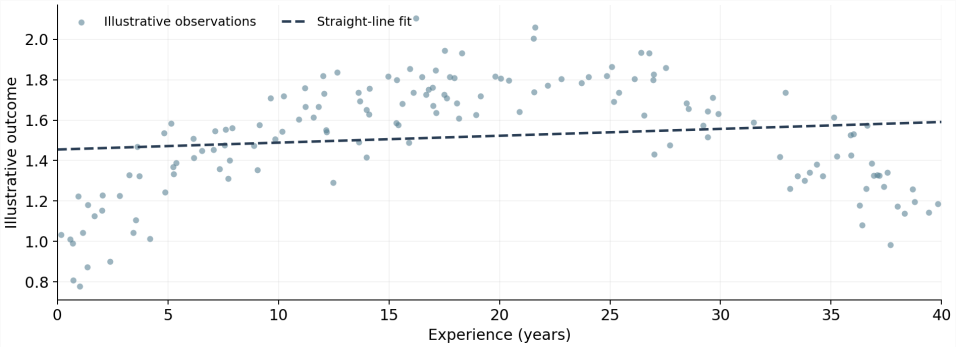
Taking logs: interpreting the slope

Form	Dependent variable	Explanatory variable	Interpretation of β_1
Level-level	y	x	One more unit of x: β_1 units of y.
Level-log	y	$\log(x)$	1% more x: about $\beta_1/100$ units of y.
Log-level	$\log(y)$	x	One more unit of x: about $100\beta_1$% in y.
Log-log	$\log(y)$	$\log(x)$	1% more x: about β_1% in y.

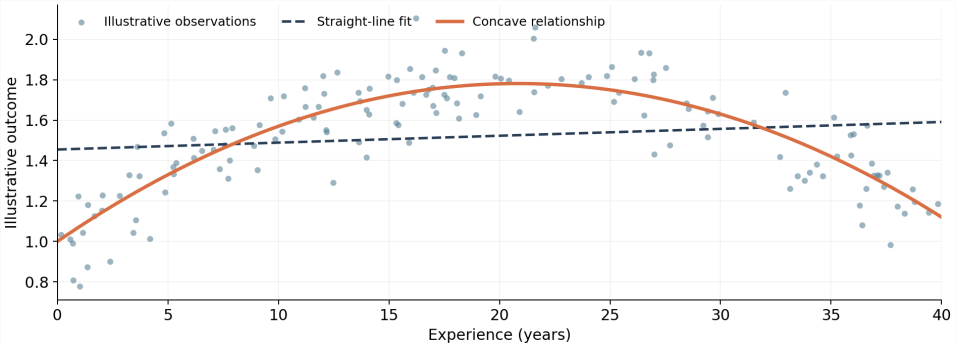
Practical choices when using logarithms

- When $y > 0$, using $\log(y)$ can help satisfy the classical linear model assumptions more closely by mitigating heteroskedasticity and skewness in the conditional distribution.
- Taking logs often narrows a variable's range. This can help with large monetary values and make OLS estimates less sensitive to outliers.
- A log requires a strictly positive value. For a proportion close to zero, taking logs can create extreme values rather than tame them.
- Rules of thumb:
  - Positive dollar amounts often enter in logs.
  - Large integer values, such as population or employees, often enter in logs.
  - Variables measured in years usually enter in levels.
  - Proportions and rates tend to enter in levels.

Models with quadratics



Models with quadratics



$$y_i = \beta_0 + \beta_1 exper_i + \beta_2 exper_i^2 + u_i.$$

Quadratic functions can be used to capture decreasing or increasing marginal effects.

The wage example

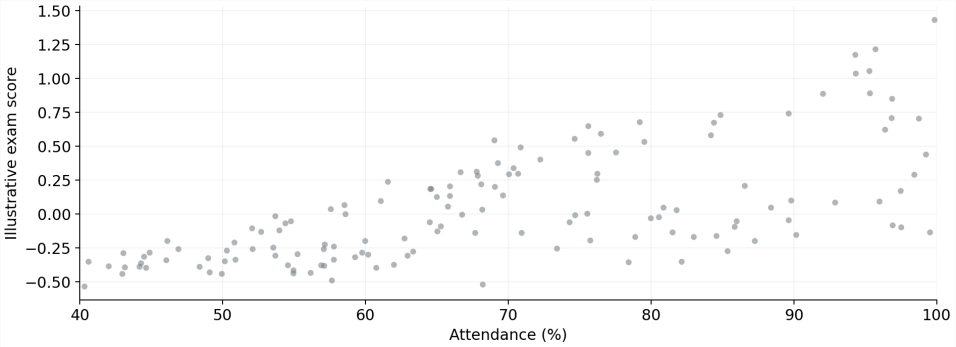
$$\widehat{wage} = 3.73 + 0.298\, exper - 0.0061\, exper^2, \qquad n = 526, \quad R^2 = 0.093.$$

$$\frac{\partial \widehat{wage}}{\partial exper} = 0.298 - 2(0.0061) exper = 0.298 - 0.0122\, exper.$$

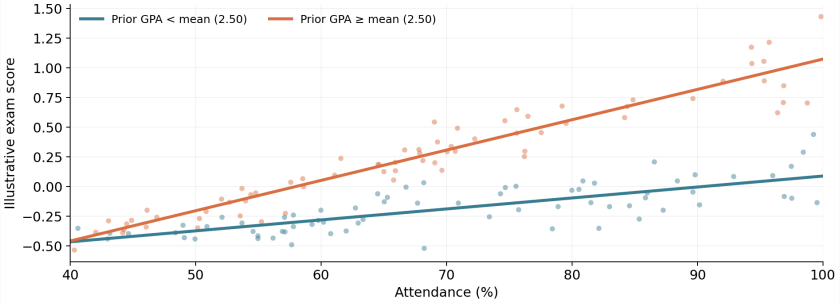
At $exper = 10$, the fitted marginal association is about \$0.176 per hour per year (17.6 cents).

- **Turning point:** $\partial \widehat{wage}/\partial exper > 0$ if $exper < exper^* = 24.4$; < 0 if $exper > exper^*$.
- **Possible explanations:** (i) Few workers beyond 24 years—but 28% are; (ii) a truly negative effect after 24 years—perhaps too early; (iii) omitted factors or an incorrect functional form distort the fit.
- **Same signs:** β_1 and β_2 could have the same sign.

Attendance and exam scores



Models with interaction terms



$$stndfnl_i = \beta_0 + \beta_1 atndrte_i + \beta_2 priGPA_i + \beta_3 (atndrte_i \times priGPA_i) + \beta_4 ACT_i + u_i.$$

The colors split GPA at its mean; the model uses continuous GPA and slope $\beta_1 + \beta_3 priGPA$.

Interaction models: one slope depends on another variable

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 z_i + \beta_3 x_i z_i + u_i.$$

$$\frac{\partial \mathbb{E}(y_i \mid x_i, z_i)}{\partial x_i} = \beta_1 + \beta_3 z_i \qquad \text{if } \mathbb{E}(u_i \mid x_i, z_i) = 0.$$

- β_3 is the change in the x slope when z rises by one unit.
- Evaluate the slope at meaningful z values, such as the mean or quartiles.
- β_1 is the x slope at $z = 0$, which may be irrelevant.
- Center z at $\bar{z}$ to make the x coefficient its slope at mean z.

Application: attendance and exam performance

Regress standardized final exam score on attendance, prior GPA, ACT, quadratic terms and an interaction term.

$$\widehat{stndfnl} = 2.05 - 0.0067\, atndrte - 1.63\, priGPA - 0.128\, ACT + 0.296\, priGPA^2 + 0.0045\, ACT^2 + 0.0056\, priGPA \cdot atndrte.$$

$$\frac{\Delta \widehat{stndfnl}}{\Delta atndrte} = -0.0067 + 0.0056\, priGPA.$$

At the sample mean $priGPA = 2.59$, one percentage point more attendance is associated with 0.0078 standard deviations higher fitted exam score.

The coefficient -0.0067 describes the slope at $priGPA = 0$, outside the sample's useful range.

More on Goodness-of-Fit and Selection of Regressors

What a small R^2 means

$$R^2 = 1 - \frac{\text{SSR}}{\text{SST}}.$$

R^2 estimates how much variation in y is explained by $\boldsymbol{x}$ in the population.

- A small R^2 means much variation in y remains unexplained: the model has not accounted for many factors that affect y.
- It does not mean those factors are correlated with the regressors. Whether u is correlated with $\boldsymbol{x}$ is a separate question.
- A small R^2 implies that error variance is large relative to the variation in y, which can reduce estimation precision.

Adjusted R^2 and extra regressors

With k independent variables and an intercept ($k+1$ coefficients),

$$\bar{R}^2 = 1 - \frac{\text{SSR}/(n-k-1)}{\text{SST}/(n-1)}.$$

Adjusted R^2 imposes a penalty for adding additional independent variables to a model. Unlike ordinary R^2, it can fall when a regressor is added.

If one independent variable is added, $\bar{R}^2$ increases if, and only if, the new variable's t statistic is greater than one in absolute value (using the same outcome and sample).

Using adjusted R^2 to choose between nonnested models

Consider choosing between the models

$$M_A: \quad y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + u_i,$$
$$M_B: \quad y_i = \gamma_0 + \gamma_1 x_{1i} + \gamma_2 x_{3i} + v_i.$$

These models are nonnested because neither is a special case of the other.

Adjusted R^2 can help compare nonnested models for the *same outcome and observations*. If x_2 and x_3 are similar or correlated, prefer the specification with the higher adjusted R^2 on fit grounds.

More than one term may differ: compare a model for y using $\log(x)$ with one using x and x^2. But do not compare models with y versus $\log(y)$ as outcomes: they have different total variation to explain.

Controlling for too many factors in regression analysis

A higher beer tax may reduce beer consumption and drunk driving, thereby reducing traffic fatalities. Start with

$$fatalities_i = \beta_0 + \beta_1 tax_i + \beta_2 miles_i + \beta_3 percmale_i + \beta_4 \texttt{perc16_21}_i + u_i.$$

Should per capita beer consumption be another regressor? Not if the aim is the total effect of tax, including its effect through drinking.

If it is included, the model becomes

$$fatalities_i = \gamma_0 + \gamma_1 tax_i + \gamma_2 beercons_i + \cdots + v_i.$$

Then γ_1 compares different tax levels *holding beer consumption fixed*; it excludes the channel through which tax changes consumption.

Adding regressors to reduce the error variance

Adding a regressor may exacerbate multicollinearity. On the other hand, removing a factor from the error generally lowers error variance. We cannot usually tell which effect dominates precision.

Under the usual homoskedasticity assumptions,

$$\mathrm{Var}(\hat{\beta}_j \mid X) = \frac{\sigma^2}{\sum_i (x_{ij} - \bar{x}_j)^2 (1 - R_j^2)}.$$

One clear case: include variables that affect y and are uncorrelated with all regressors of interest.

- Adding such a variable does not exacerbate multicollinearity in the population.
- It reduces the error variance.

Residual Analysis

Why inspect residuals?

Before any tests or interpretation of results, check whether the empirical evidence contradicts the maintained model assumptions. We do this by analyzing the residuals.

Write X for the $n \times (k+1)$ regressor matrix. If $y = X\beta + u$, then

$$\hat{u} = y - X\hat{\beta} = (I - P_X)y = (I - P_X)u, \qquad P_X = X(X'X)^{-1}X'.$$

P_X projects y onto the fitted values: $P_X^2 = P_X$ and $\mathrm{tr}(P_X) = k+1$. Since $\hat{y} = P_X y$,

$$\hat{y}_i = \sum_{j=1}^{n} P_{ij} y_j, \qquad P_{ii} = x_i'(X'X)^{-1}x_i.$$

P_{ii} is the weight of y_i in its own fitted value—its *leverage*. With an intercept and one regressor,

$$P_{ii} = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{\sum_j (x_j - \bar{x})^2}.$$

Three ways to scale a residual

The raw residual $\hat{u}_i$ is in units of y. Scaling makes residuals easier to compare:

$$\text{Standardized:} \quad r_i = \frac{\hat{u}_i}{\hat{\sigma}},$$

$$\text{Internally studentized:} \quad t_i = \frac{\hat{u}_i}{\hat{\sigma}\sqrt{1 - P_{ii}}},$$

$$\text{Externally studentized:} \quad t_i^* = \frac{\hat{u}_i}{\hat{\sigma}_{(i)}\sqrt{1 - P_{ii}}}.$$

Standardizing removes units but ignores that high-leverage residuals have smaller variance. Internal studentization divides by their estimated standard deviation. External studentization also estimates σ without case i, so that case cannot inflate its own denominator.

Why leverage enters the residual scale

Assume fixed, full-rank X and independent normal errors $u \sim N(0, \sigma^2 I)$. Because $I - P_X$ is a projection,

$$\hat{u} \sim N\big(0, \sigma^2 (I - P_X)\big), \qquad \operatorname{Var}(\hat{u}_i \mid X) = \sigma^2 (1 - P_{ii}).$$

Consequently,

$$\frac{\hat{u}_i}{\sigma \sqrt{1 - P_{ii}}} \sim N(0, 1).$$

Holding X fixed, $\partial \hat{y}_i / \partial y_i = P_{ii}$ and $\partial \hat{u}_i / \partial y_i = 1 - P_{ii}$. Thus a high-leverage point pulls its own fitted value toward y_i and can have a deceptively small raw residual.

The externally studentized residual

Under the same normal, homoskedastic model, the left-out variance estimate is independent of $\hat{u}_i$ and

$$\frac{(n-k-2)\hat{\sigma}^2_{(i)}}{\sigma^2} \sim \chi^2_{n-k-2}.$$

It follows that

$$t_i^* = \frac{\hat{u}_i}{\hat{\sigma}_{(i)}\sqrt{1-P_{ii}}} \sim t_{n-k-2}.$$

For large n, this is close to $N(0,1)$. A large $|t_i^*|$ flags a case that is unusual under the model. But if we inspect many cases, a few large values can occur by chance even when the model holds.

What the residuals can check

Look for visible conflicts with the assumptions of the maintained model:

- **Zero conditional mean:** for the chosen functional form, no systematic residual pattern against fitted values or regressors.
- **Random sampling:** if observations have a meaningful sequence, such as collection date, residual runs or cycles may suggest dependence. Spreadsheet row numbers can be shuffled, so they tell us nothing about dependence.
- **Homoskedasticity:** residual spread should not change systematically with the fitted value.
- **Normality:** inspect the distribution of studentized residuals when using exact small-sample inference.

With an intercept, $\sum_i \hat{u}_i = 0$ by construction; this does not establish zero conditional mean in the population.

A plot for each diagnostic question

Plot	What would raise concern?
$\hat{u}_i$ versus $\hat{y}_i$ or an x_j	Curvature or another systematic mean pattern
$\lvert\hat{u}_i\rvert$ or $\hat{u}_i^2$ versus $\hat{y}_i$	Spread that changes with the fitted value
Residuals by time or collection order	Runs, cycles or abrupt shifts; arbitrary row order has no meaning
Studentized residual histogram and normal Q–Q plot	Skew or unusually heavy tails

Plots identify questions for follow-up; they do not by themselves establish the cause of a model failure.

A running example: wage and experience

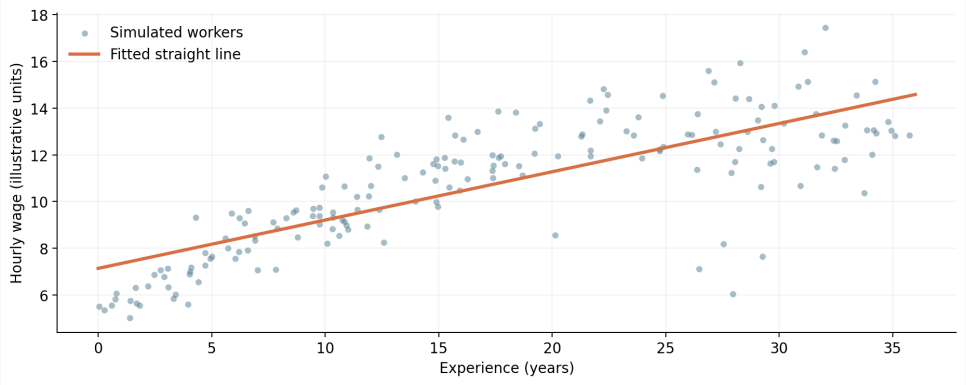
Consider 180 simulated workers. Start with the simple specification

$$wage_i = \beta_0 + \beta_1 exper_i + u_i.$$

The fitted line asks whether hourly wage changes at a constant rate as experience rises. We will inspect its residuals before interpreting its slope.

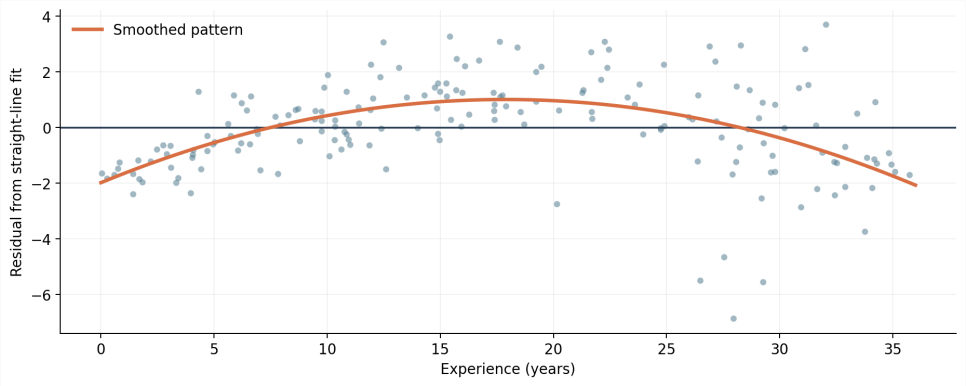
The same observations will be used for every diagnostic on the next slides. The example is simulated to make the patterns visible.

Example: the fitted straight line



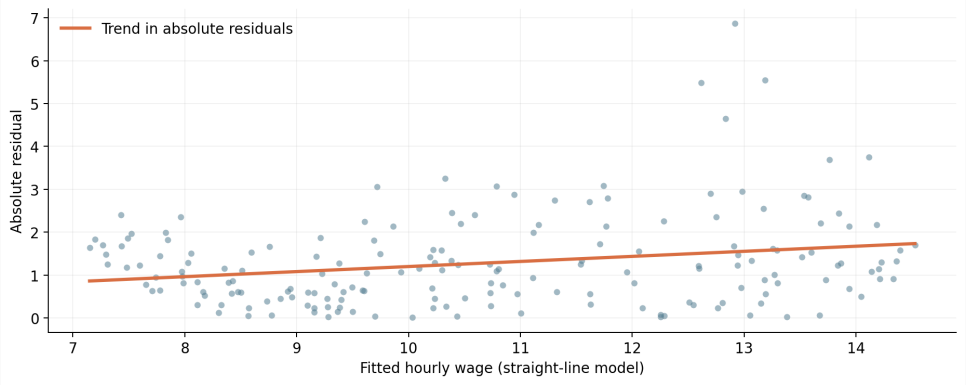
At higher experience, many wages lie below the fitted line. This suggests the relationship may bend rather than follow a constant slope.

Example: residuals reveal curvature



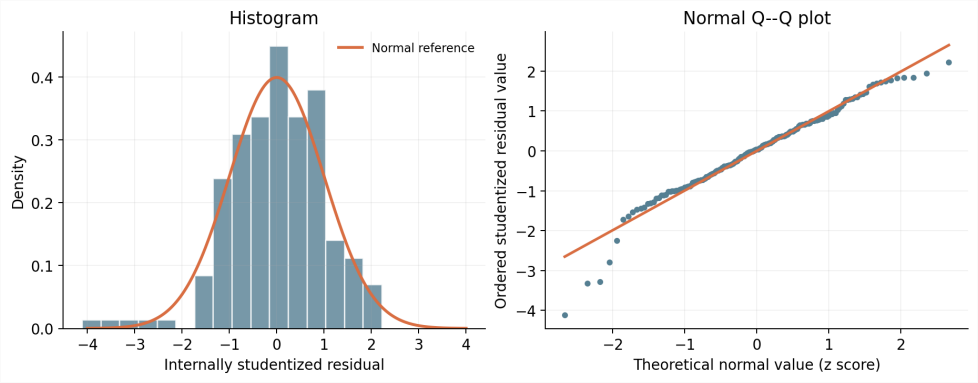
Residuals tend to be negative at the ends of the experience range and positive in the middle. A straight line misses this concavity.

Example: residual spread changes



Absolute residuals tend to grow with the fitted wage. This suggests nonconstant error variance, a topic we will examine in Lecture 6.

Example: inspect the residual distribution



The Q–Q plot pairs ordered residuals with normal z-scores. Its horizontal values (such as -2) are normal quantile *values*, not probabilities. Percentile probabilities range from 0 to 1.

Cook's distance measures influence

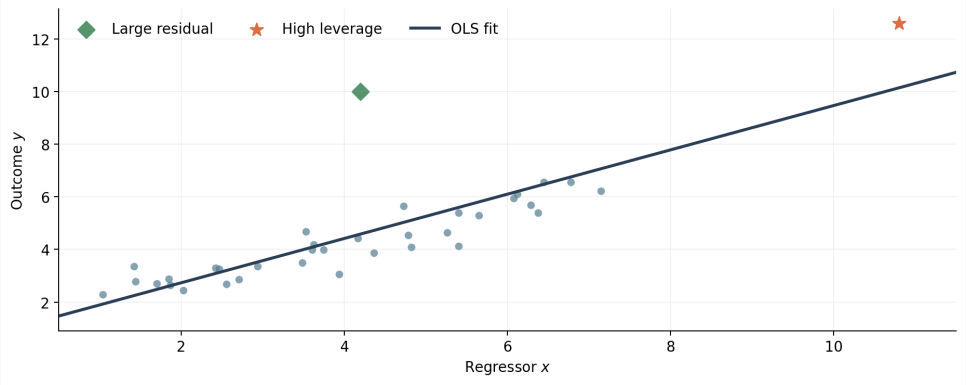
Let $\hat{y}_j$ be the fitted value from OLS estimated on all n observations. Let $\hat{y}_{j(i)}$ be the fitted value at the same x_j after re-estimating OLS without observation i. Then

$$D_i = \frac{\sum_{j=1}^{n} (\hat{y}_j - \hat{y}_{j(i)})^2}{(k+1)\hat{\sigma}^2}.$$

Cook's distance summarizes how much one observation changes the fitted values. It combines residual size and leverage; neither alone is the same as influence.

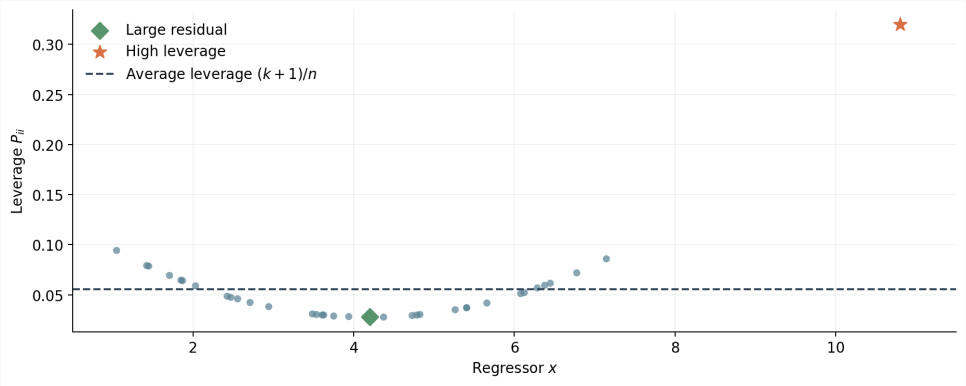
A large value prompts a check of the observation's source, coding and place in the regressor range.

Example: two unusual observations



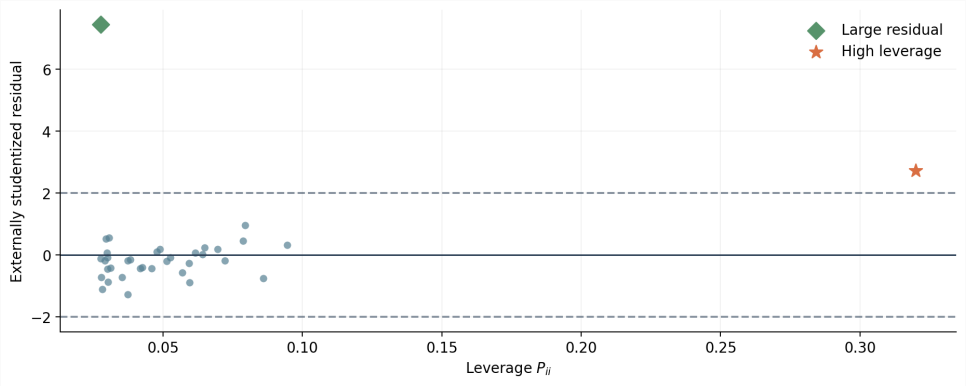
The green point has an unusually large vertical gap. The orange point lies far from the other x values. We need separate measures for these two features.

Example: leverage is about the regressors



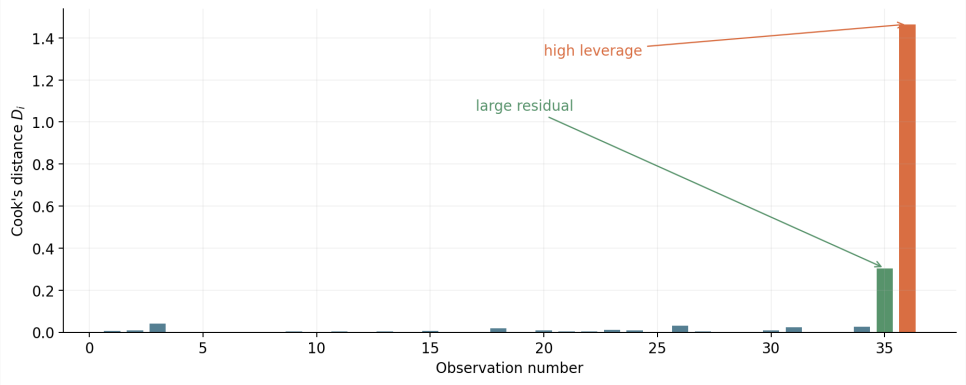
P_{ii} depends only on X; its average is $(k+1)/n$. The orange point has high leverage because its x value is far from the bulk of the sample.

Example: residual size and leverage together



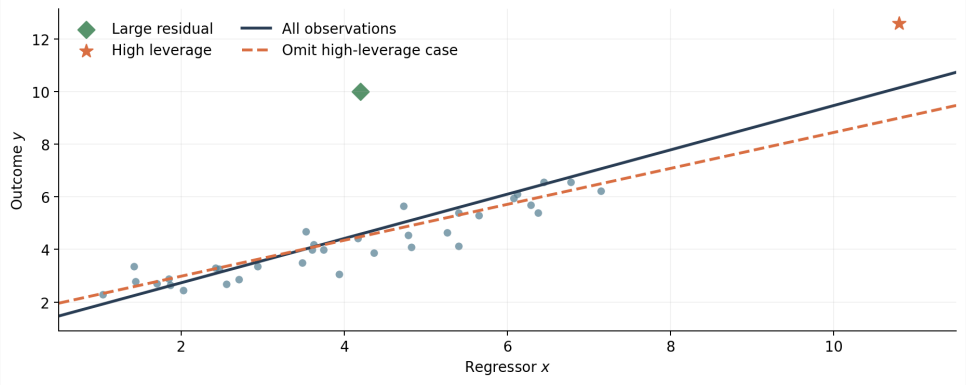
The green point is unusual in y given its x; the orange point is unusual in x. Their influence on the estimated line remains a separate empirical question.

Example: compare Cook's distances



Cook's distance ranks observations by how much the fitted model changes if each one is removed. Investigate the largest values; a cutoff is a screening aid, not an automatic deletion rule.

Example: refit without an influential case



The change in fitted line makes the influence visible. Report the full-sample estimate and the sensitivity result, then decide whether the observation represents the target population.

Outliers and influential observations

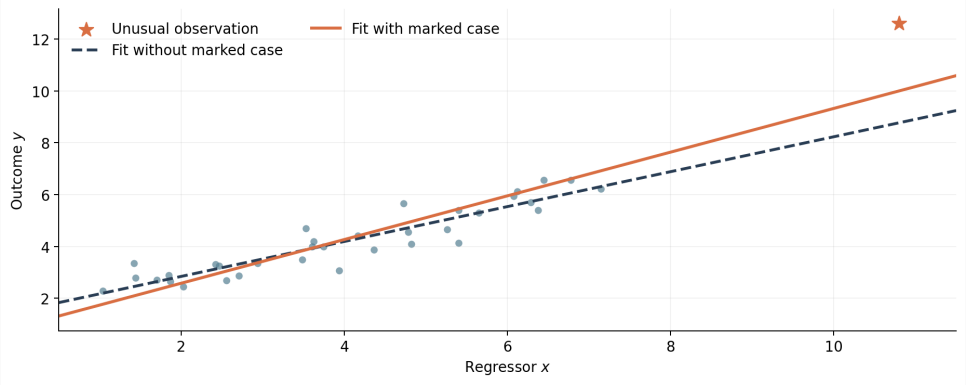
OLS can change substantially when a few unusual observations drive the sum of squared residuals.

For a single variable, let Q_1 and Q_3 be its first and third quartiles (25th and 75th percentiles). Tukey's exploratory fences are

$$\begin{aligned} &\text{Unusual:} \quad x < Q_1 - 1.5(Q_3 - Q_1) \quad \text{or} \quad x > Q_3 + 1.5(Q_3 - Q_1), \\ &\text{Extreme:} \quad x < Q_1 - 3(Q_3 - Q_1) \quad \text{or} \quad x > Q_3 + 3(Q_3 - Q_1). \end{aligned}$$

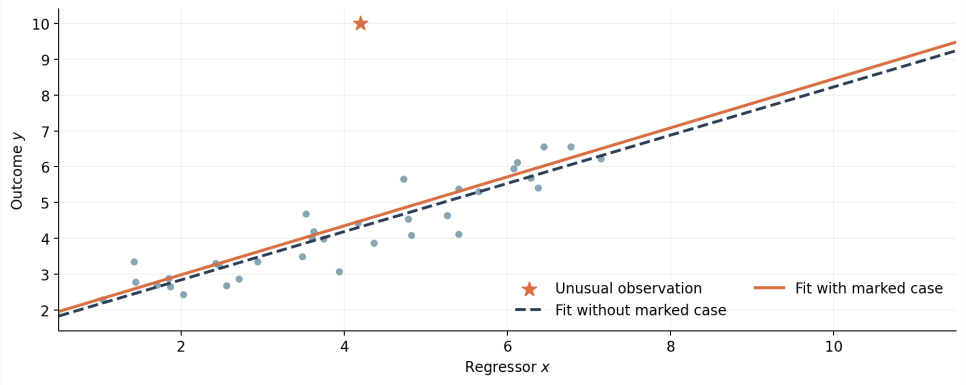
These fences describe a marginal distribution. In regression, also distinguish a large residual, high leverage and influence on the fit.

An unusual regressor value can move the line



The marked observation lies far from the other x values. Its leverage gives it the potential to change the slope markedly, even if its residual from the full fit is small.

An unusual outcome has a large residual



The marked observation has an unusual y given its ordinary x value. It can matter to OLS, but its location gives it less leverage over the slope than a far-away x value.

How to investigate a flagged observation

- **Large residual:** inspect externally studentized $|t_i^*|$; for large n, 1.96 is an approximate 5% flag for one case.
- **High leverage:** inspect P_{ii} relative to its average $(k+1)/n$. If P_{ii} is near 1, $\sqrt{1-P_{ii}}$ is small, so a modest raw residual can produce a large $|t_i^*|$.
- **Influence:** use Cook's distance, then check coding, units and whether the case belongs to the target population. High leverage alone does not imply influence.
- Correct a verified error. For a valid but influential case, report sensitivity and consider a method that limits undue influence.

Why OLS is sensitive to large residuals

OLS minimizes squared residuals:

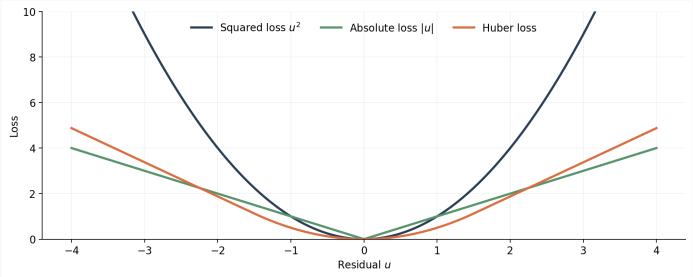
$$\hat{\beta}_{OLS} = \arg\min_b \sum_{i=1}^{n} (y_i - x_i' b)^2.$$

For one residual u, the loss is u^2 and its slope is $2u$. Thus a large residual can dominate the first-order condition:

$$\sum_{i=1}^{n} x_{ij} \hat{u}_i = 0 \qquad (j = 0, \ldots, k).$$

This explains why a few extreme observations can move the fitted coefficients substantially.

Less sensitive loss functions



Absolute loss $|u|$ gives bounded weight to an extreme residual. Huber loss is quadratic near zero and linear beyond a:

$$\rho_a(u) = \begin{cases} \frac{1}{2}u^2, & |u| \le a, \\ a(|u| - \frac{1}{2}a), & |u| > a. \end{cases}$$

Trimming and winsorization

To reduce the effect of extreme residuals, use this sensitivity procedure:

- Fit OLS and calculate the residuals. Identify the bottom and top 2.5% (5% in total).
- **Trimming:** omit those observations. **Winsorization:** keep them, but replace an extreme y_i by $\hat{y}_i$ plus the 2.5th or 97.5th percentile residual, as appropriate.
- Re-estimate OLS on the reduced or modified sample. Winsorization can be iterated.

These procedures can bias second-moment estimates and change the target of estimation. State the rule and report the original OLS result beside the sensitivity result.

Chapter 7
Multiple Regression Analysis with Qualitative Information: Binary (or Dummy) Variables

Qualitative information as indicators

An indicator (or dummy) variable equals one when an observation satisfies a stated condition, and zero otherwise.

Examples include whether a person received a treatment, has a supplementary retirement plan, or uses a computer at work.

The interpretation starts with the coding: name the group represented by one and the reference group represented by zero.

One indicator produces two parallel regression lines

Consider

$$y_i = \beta_0 + \beta_1 x_i + \delta D_i + u_i, \qquad D_i \in \{0, 1\}.$$

If $\mathbb{E}(u_i \mid x_i, D_i) = 0$, the conditional mean is

$$\begin{aligned} D_i = 0: &\quad \mathbb{E}(y_i \mid x_i, D_i = 0) = \beta_0 + \beta_1 x_i, \\ D_i = 1: &\quad \mathbb{E}(y_i \mid x_i, D_i = 1) = (\beta_0 + \delta) + \beta_1 x_i. \end{aligned}$$

The lines have the same slope. At a given x, δ is their vertical difference. If the outcome is in logs, interpret the group difference in proportional terms.

Changing the reference group

Let $E_i = 1 - D_i$, so $E_i = 1$ identifies the group previously coded $D_i = 0$. Substitute $D_i = 1 - E_i$:

$$\begin{aligned} y_i &= \beta_0 + \beta_1 x_i + \delta(1 - E_i) + u_i \\ &= (\beta_0 + \delta) + \beta_1 x_i - \delta E_i + u_i. \end{aligned}$$

The new intercept is $\beta_0^* = \beta_0 + \delta$ and the new indicator coefficient is $\delta^* = -\delta$.

Only the reference and sign of the reported contrast change; fitted values are identical.

Qualitative variables with several categories

Many qualitative variables have more than two possible values:

- marital status can have several distinct categories;
- a survey response may range from strong agreement to strong disagreement;
- a geographic variable may identify several regions.

These labels describe categories. Treating their numeric codes as a single quantitative regressor would impose distances between categories that the labels need not support.

For a low/middle/high rating, middle and high indicators (with low as the reference) allow two different effects; coding the levels $1, 2, 3$ forces equal steps.

For an ordinal variable, separate indicators can retain the ordering without assuming equally spaced effects.

Partition the observations into categories

Let Ω denote the observations under study. The categories $C_1, \ldots, C_S$ are mutually exclusive and exhaustive:

$$\bigcup_{s=1}^{S} C_s = \Omega, \qquad C_s \cap C_r = \varnothing \quad (s \neq r).$$

Create one indicator for each category:

$$D_{si} = \begin{cases} 1, & i \in C_s, \\ 0, & i \notin C_s. \end{cases}$$

Every observation belongs to exactly one category, so $\sum_{s=1}^{S} D_{si} = 1$ for every i.

The full set of category indicators

With one continuous regressor, a model containing all S indicators is

$$y_i = \beta_0 + \beta_1 x_i + \sum_{s=1}^{S} \delta_s D_{si} + u_i.$$

Writing D_s for the $n \times 1$ vector of category-s indicators,

$$y = Z\theta + u, \qquad Z = \begin{bmatrix} 1_n & x & D_1 & \cdots & D_S \end{bmatrix}, \qquad \theta = (\beta_0, \beta_1, \delta_1, \ldots, \delta_S)'.$$

Z has $S + 2$ columns: an intercept, x, and one column for each category. The next question is whether all these coefficients can be separately estimated.

Why the full indicator model is not identified

Because each observation is in one category, $\sum_s D_{si} = 1$. For any number λ,

$$(\beta_0 - \lambda) + \sum_{s=1}^{S} (\delta_s + \lambda) D_{si} = \beta_0 + \sum_{s=1}^{S} \delta_s D_{si} + \lambda \left(\sum_{s=1}^{S} D_{si} - 1 \right)$$
$$= \beta_0 + \sum_{s=1}^{S} \delta_s D_{si}.$$

Thus shifting the intercept down by λ and every category coefficient up by λ gives exactly the same fitted value for every observation. The model is not identified: $(\beta_0, \beta_1, \delta_1, \ldots, \delta_S)$ and $(\beta_0 - \lambda, \beta_1, \delta_1 + \lambda, \ldots, \delta_S + \lambda)$ are two possible parametrizations of the same regression function.

The sum of squared residuals is flat in one direction

OLS minimizes the objective

$$\min_{\beta_0,\beta_1,\delta_1,\ldots,\delta_S} \mathrm{SSR}(\beta_0,\beta_1,\delta_1,\ldots,\delta_S) = \min_{\beta_0,\beta_1,\delta_1,\ldots,\delta_S} \sum_{i=1}^{n} \left(y_i - \beta_0 - \beta_1 x_i - \sum_{s=1}^{S} \delta_s D_{si} \right)^2 .$$

The parameter shift on the preceding slide leaves every residual unchanged. Therefore, for every $\lambda \in \mathbb{R}$,

$$\mathrm{SSR}(\beta_0 - \lambda, \beta_1, \delta_1 + \lambda, \ldots, \delta_S + \lambda) = \mathrm{SSR}(\beta_0, \beta_1, \delta_1, \ldots, \delta_S).$$

If one set of coefficients minimizes SSR, infinitely many shifted sets minimize it too. The problem concerns the separate coefficients, even though the fitted values are the same.

The same problem as a rank deficiency

In the design matrix $Z = [1_n, x, D_1, \ldots, D_S]$,

$$1_n = \sum_{s=1}^{S} D_s.$$

One column is a linear combination of the others, so Z does not have full column rank. Consequently $Z'Z$ is singular and the usual OLS inverse $(Z'Z)^{-1}$ does not exist.

This is the dummy-variable trap. It is the matrix version of the coefficient shift shown on the preceding slides.

An identifying restriction

To estimate a unique set of category coefficients, remove the redundant degree of freedom. Two common choices are

$$\delta_{s_0} = 0 \qquad \text{or} \qquad \sum_{s=1}^{S} \delta_s = 0.$$

The first uses category s_0 as the reference. The second measures each effect relative to the average category intercept.

Once the remaining regressors have full rank, different permitted coefficient vectors give different fitted values and OLS has a unique coefficient solution.

The restrictions change the labels attached to intercepts and category effects; they need not change the fitted values of the model.

An identifying restriction: omit one category as the reference

Choose category s_0 as the reference and **omit its indicator**, setting $\delta_{s_0} = 0$:

$$y_i = \beta_0 + \beta_1 x_i + \sum_{s \neq s_0} \delta_s D_{si} + u_i.$$

Under zero conditional mean, the reference-group intercept is β_0. For group $s \neq s_0$, it is $\beta_0 + \delta_s$; hence δ_s compares group s with the reference at the same x.

For example, a marital-status model may use single people as the reference. Changing the reference changes reported coefficients but not fitted values.

Alternative identifying restriction: effects that sum to zero

Keep all S indicators and impose $\sum_{s=1}^{S} \delta_s = 0$. Each category has intercept $\beta_0 + \delta_s$.

Here β_0 is the unweighted average of the S category intercepts, and δ_s is that category's departure from this average. This differs from choosing one category as the reference.

Remark: With an intercept, only $S - 1$ category effects are free.
A category absent from the sample has no estimable effect.

Alternatively, omit the intercept and keep all S indicators. Each indicator coefficient is then its category's intercept, provided that category appears in the sample.

Several categorical variables in one regression

Suppose we observe four regions and three education groups. With an intercept, include three region indicators and two education indicators:

$$y_i = \beta_0 + \beta_1 x_i + \sum_{r=2}^{4} \rho_r R_{ri} + \sum_{e=2}^{3} \eta_e E_{ei} + u_i.$$

Each coefficient compares its category with the omitted category of the *same* variable, holding the other variables fixed.

This additive model gives the same region contrast in every education group. Add region–education interactions if those contrasts may differ by education.

A dummy variable when the outcome is in logs

For $y_i > 0$, consider

$$\mathbb{E}(\log y_i \mid x_i, D_i) = \beta_0 + \beta_1 x_i + \delta D_i.$$

At fixed x, changing D from zero to one changes the conditional mean of $\log y$ by δ.

Exponentiating the fitted log values, the proportional difference is

$$\frac{\exp(\beta_0 + \beta_1 x + \delta) - \exp(\beta_0 + \beta_1 x)}{\exp(\beta_0 + \beta_1 x)} = e^{\delta} - 1.$$

Interpret the coefficient as approximately $100\delta\%$; the exact change in exponentiated fitted log values is $100(e^{\delta} - 1)\%$.

Application: training grants and training hours (Example 7.3)

For 105 Michigan manufacturing firms:

$$\widehat{hrsemp} = 46.67 + 26.25\, grant - 0.98 \log(sales) - 6.07 \log(employ).$$

Here *hrsemp* is annual training hours per employee; $grant = 1$ means the firm received a training grant.

Controlling for sales and employment, firms that received a grant trained each worker, on average, 26.25 hours more.

hrsemp ranges from 0 to 164 hours, with a sample average of about 17 hours.

Application: colonial style and house prices (Example 7.4)

For 88 houses:

$$\begin{aligned}\widehat{\log(price)} = &-1.35 + 0.168\log(lotsize) + 0.707\log(sqrft) \\ &+ 0.027\,bdrms + 0.054\,colonial.\end{aligned}$$

$colonial = 1$ denotes a colonial-style house; other houses form the reference group.

A colonial-style house is predicted to sell for about 5.4% more, holding other factors fixed.

The approximate difference is **5.4%**; the exact difference in exponentiated fitted log prices is $100(e^{0.054} - 1) \approx$ **5.6%**.

A category can change both intercept and slope

Introduce an interaction between an indicator and a continuous regressor:

$$y_i = \beta_0 + \beta_1 x_i + \delta_0 D_i + \delta_1 D_i x_i + u_i.$$

If $\mathbb{E}(u_i \mid x_i, D_i) = 0$, the two conditional mean lines are

$$\begin{aligned} D_i = 0: &\quad \mathbb{E}(y_i \mid x_i, D_i = 0) = \beta_0 + \beta_1 x_i, \\ D_i = 1: &\quad \mathbb{E}(y_i \mid x_i, D_i = 1) = (\beta_0 + \delta_0) + (\beta_1 + \delta_1) x_i. \end{aligned}$$

δ_0 is the intercept difference and δ_1 is the slope difference. The effect of a change Δx is $(\beta_1 + \delta_1 D_i)\Delta x$ within group D_i.

When separate group regressions give the same OLS fit

Write $\beta_0^* = \beta_0 + \delta_0$ and $\beta_1^* = \beta_1 + \delta_1$. The pooled OLS objective becomes

$$\sum_{i:D_i=0} (y_i - \beta_0 - \beta_1 x_i)^2 + \sum_{i:D_i=1} (y_i - \beta_0^* - \beta_1^* x_i)^2.$$

The two sums involve disjoint groups and disjoint parameters. Minimizing them together therefore gives the same fitted lines as two separate regressions, one in each group.

This equivalence requires a fully group-specific intercept and slope. If a coefficient is constrained to be common across groups, estimate the model on the pooled sample.

Application: does the education slope differ by gender? (Example 7.10)

Wooldridge estimates, using 526 workers,

$$\widehat{\log(wage)} = 0.389 - 0.227\,female + 0.082\,educ - 0.0056\,(female \times educ) + 0.029\,exper - 0.00058\,exper^2 + 0.032\,tenure - 0.00059\,tenure^2.$$

An extra year of education is associated with about **8.2%** higher wages for men and $(0.082 - 0.0056) \times 100 = \mathbf{7.6\%}$ for women, holding experience and tenure fixed.

A Binary Dependent Variable: The Linear Probability Model

The linear probability model

Let $y_i \in \{0, 1\}$, where $y_i = 1$ denotes success. Consider the usual model

$$y_i = \beta_0 + \sum_{j=1}^{k} \beta_j x_{ji} + u_i.$$

Under zero conditional mean, $\mathbb{E}(u_i \mid \boldsymbol{x}_i) = 0$, and because y_i is binary,

$$\mathbb{E}(y_i \mid \boldsymbol{x}_i) = \beta_0 + \sum_{j=1}^{k} \beta_j x_{ji} = \Pr(y_i = 1 \mid \boldsymbol{x}_i) \equiv p(\boldsymbol{x}_i).$$

The probability of success $p(\boldsymbol{x})$ is a linear function of the x_j. Holding other factors fixed, β_j measures its change when x_j changes:

$$\Delta \Pr(y_i = 1 \mid \boldsymbol{x}_i) = \beta_j \Delta x_{ji}.$$

For $\hat{y}_i = \hat{\beta}_0 + \sum_{j=1}^{k} \hat{\beta}_j x_{ji}$, $\hat{y}_i$ is the predicted success probability. $\hat{\beta}_0$ is the prediction at all $x_j = 0$; $\hat{\beta}_j$ gives the predicted change from a one-unit increase in x_j.

Application: married women's labor-force participation (7.29)

In a 1975 sample of 753 married women, $inlf = 1$ if the woman worked for pay during the year (428 did so):

$$\widehat{inlf} = 0.586 - 0.0034\, nwifeinc + 0.038\, educ + 0.039\, exper \\ - 0.00060\, exper^2 - 0.016\, age - 0.262\, kidslt6 + 0.013\, kidsge6.$$

nwifeinc is non-wife income in thousands of 1975 dollars; *kidslt*6 and *kidsge*6 count children under six and aged 6–18.

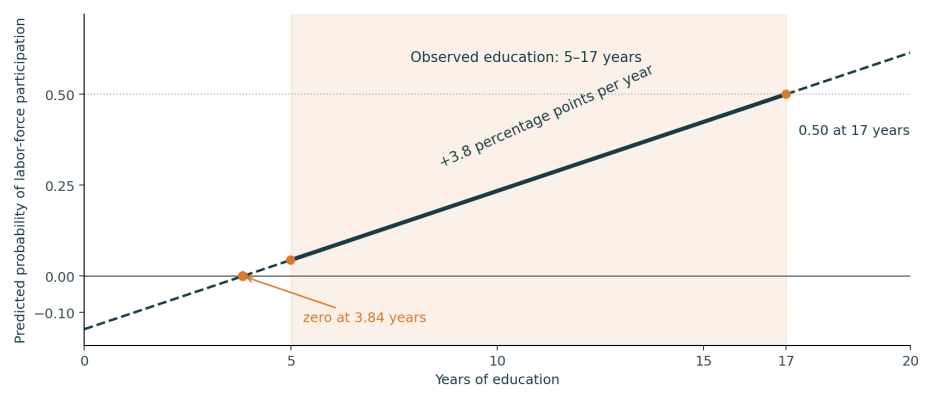
One more year of education is associated with a **3.8-percentage-point higher** predicted participation probability, holding the other variables fixed.

One more child under six reduces the predicted probability of participation by 0.262 (coefficient -0.262), holding the other variables fixed.

Education and predicted labor-force participation

In (7.29), set $nwifeinc = 50$, $exper = 5$, $age = 30$, $kidslt6 = 1$ and $kidsge6 = 0$:

$$\widehat{\Pr(inlf = 1 \mid \boldsymbol{x})} = -0.146 + 0.038\, educ.$$



Linear probability model: limits and uses

- Fitted probabilities can be negative or exceed one. A probability cannot be linear in the regressors over all possible values.
- The model is still useful near sample averages, where fitted probabilities are often plausible.
- To classify outcomes, set $\tilde{y}_i = 1$ if $\hat{y}_i \geq 0.5$, and $\tilde{y}_i = 0$ otherwise.
- For $p(\boldsymbol{x}) = \Pr(y = 1 \mid \boldsymbol{x})$,

$$\operatorname{Var}(u \mid \boldsymbol{x}) = p(\boldsymbol{x})[1 - p(\boldsymbol{x})].$$

  Unless $p(\boldsymbol{x})$ is constant, this variance generally depends on $\boldsymbol{x}$: the model is heteroskedastic. Use robust standard errors for inference.

Interpreting Regression Results with Discrete Dependent Variables

A count slope describes an expected count

Examples of discrete dependent variables include:

- the number of arrests in a year;
- the number of children;
- labor-force participation (zero or one): a binary variable is an extreme case.

In the fertility regression, each additional year of education reduces the estimated number of children by 0.09, holding age fixed.

The slope describes a change in the *expected or average* value of y, not a fractional change for one woman.

If each woman in a group of 100 obtains another year of education, we estimate there will be nine fewer children among them.

The two chapters in one empirical workflow

Specify the conditional mean for the question: units, logs, curvature, interactions and categories determine the comparison.

Check fit, prediction uncertainty, residual patterns and the support of the data.

A group difference needs a credible design or substantive argument for a causal interpretation.

Next lecture: heteroskedasticity changes the uncertainty calculation.